MATH 161: U-SUBSTITUTION PRACTICE

Name: Solutions

Find the following indefinite integrals using the substitution method. Use proper notation and state the values for u and du.

1. $\int x(2x^2+3)^4 dx$

$u = 2x^2+3$

$du = 4x\,dx \Rightarrow \frac{1}{4}du = x\,dx$

$\int x(2x^2+3)^4 dx = \frac{1}{4}\int u^4 du = \frac{1}{4}\left(\frac{1}{5}u^5\right)+C = \boxed{\frac{1}{20}(2x^2+3)^5+C}$

2. $\int \sin^2\theta\cos\theta d\theta$

$u = \sin(\theta)$

$du = \cos(\theta)d\theta$

$\int \sin^2(\theta)\cos(\theta)d\theta = \int u^2 du = \frac{1}{3}u^3+C = \frac{1}{3}\sin^3(\theta)d\theta$

3. $\int x^2\sin(x^3)dx$

$u = x^3$

$du = 3x^2 dx$

$\Rightarrow \frac{1}{3}du = x^2 dx$

$\int x^2\sin(x^3)dx = \frac{1}{3}\int \sin(u)du = \frac{-1}{3}\cos(u)+C = \boxed{\frac{-1}{3}\cos(x^3)+C}$

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4. $\int \sin t\sqrt{1+\cos t}dt$

$u = 1+\cos(t)$

$du = -\sin(t)dt \Rightarrow -du = \sin(t)dt$

$\Rightarrow \int \sin(t)\sqrt{1+\cos(t)}\,dt = \int -\sqrt{u}\,du = -\frac{2}{3}u^{3/2}+C = \boxed{-\frac{2}{3}(1+\cos(t))^{3/2}+C.}$

5. $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx$

$u = \sqrt{x}$

$du = \frac{1}{2}x^{-1/2}dx \Rightarrow 2du = \frac{dx}{\sqrt{x}}$

$\Rightarrow \int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx = 2\int \cos(u)du = 2\sin(u)+C = \boxed{2\sin(\sqrt{x})+C.}$

6. $\int \sin x \sin(\cos x)dx$

$u = \cos(x)$

$du = -\sin(x)dx$

$\Rightarrow -du = \sin(x)dx$

$\int \sin(x)\sin(\cos(x))dx = \int \sin(u)du = -\cos(u)+C = \boxed{-\cos(\cos(x))+C}$

7. $\int \frac{\sec^2 x}{\tan^2 x}dx$

$u = \tan(x)$

$du = \sec^2(x)dx$

$\int \frac{\sec^2(x)}{\tan^2(x)}dx = \int \frac{1}{u^2}du = \int u^{-2}du = -u^{-1}+C = \frac{-1}{\tan(x)}+C = \boxed{-\cot(x)+C}$

Alternatively:

$\frac{\sec^2(x)}{\tan^2(x)} = \frac{1}{\cos^2(x)}\frac{\cos^2(x)}{\sin^2(x)} = \frac{1}{\sin^2(x)} = \csc^2(x) = \frac{d}{dx}(-\cot(x))$

$\Rightarrow \int \frac{\sec^2(x)}{\tan^2(x)}dx = -\cot(x)+C.$

Evaluate the following definite integrals using substitution.

8. $\displaystyle\int_0^1 \cos\left(\frac{\pi t}{2}\right) dt$

$u = \frac{\pi t}{2}$

$du = \frac{\pi}{2} dt \Rightarrow dt = \frac{2}{\pi} du$

$$\int_0^1 \cos\left(\frac{\pi t}{2}\right) dt = \int_{u(0)}^{u(1)} \cos(u) \frac{2}{\pi} du = \frac{2}{\pi} \int_0^{\pi/2} \cos(u) du = \frac{2}{\pi}\left[\sin(\pi/2) - \sin(0)\right] = \frac{2}{\pi}\left[1 - 0\right] = \boxed{\frac{2}{\pi}}$$

9. $\displaystyle\int_0^4 \frac{x}{\sqrt{1+2x}} dx$

$u = 1 + 2x$; $x = \frac{u-1}{2}$

$du = 2dx$

$\Rightarrow \frac{1}{2} du = dx$

$u(0) = 1 + 0 = 1$

$u(4) = 1 + 8 = 9$

$$= \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}} \frac{1}{2} du = \frac{1}{4} \int_1^9 \frac{u-1}{\sqrt{u}} du = \frac{1}{4}\left[\int_1^9 u^{1/2} du - \int_1^9 u^{-1/2} du\right]$$

$$= \frac{1}{4}\left[\frac{2}{3} u^{3/2}\Big|_1^9 - 2u^{1/2}\Big|_1^9\right]$$

$$= \frac{1}{4}\left[\frac{2}{3}\left(9^{3/2} - 1^{3/2}\right) - 2\left(\sqrt{9} - \sqrt{1}\right)\right]$$

$$= \frac{1}{4}\left[\frac{2}{3}(27 - 1) - 2(3 - 1)\right]$$

$$= \frac{1}{4}\left(\frac{2}{3}(26) - 4\right)$$

$$= \frac{1}{4}\left(\frac{52 - 12}{3}\right)$$

$$= \frac{1}{12}(40)$$

$$= \frac{1}{4 \cdot 3}(4 \cdot 10) = \boxed{\frac{10}{3}}$$

10. Show the following two integrals are equivalent:

$$\int_0^2 3x\sqrt{9-x^2}dx = \int_5^9 \frac{3\sqrt{u}}{2}du.$$

$u = 9 - x^2$

$du = -2x\,dx$

$\Rightarrow -\frac{1}{2}du = x\,dx$

$$\int_0^2 3x\sqrt{9-x^2}\,dx = \int_{u(0)}^{u(2)} -3/2\sqrt{u}\,du = -\int_9^5 \frac{3}{2}\sqrt{u}\,du = \int_5^9 \frac{3\sqrt{u}}{2}du. \blacksquare$$